Solution Architecture Template

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Date	03 October 2026	**Team ID**	SWTID-2026-2670
Project Name	IT Procurement Automation	**Maximum Marks**	4 Marks

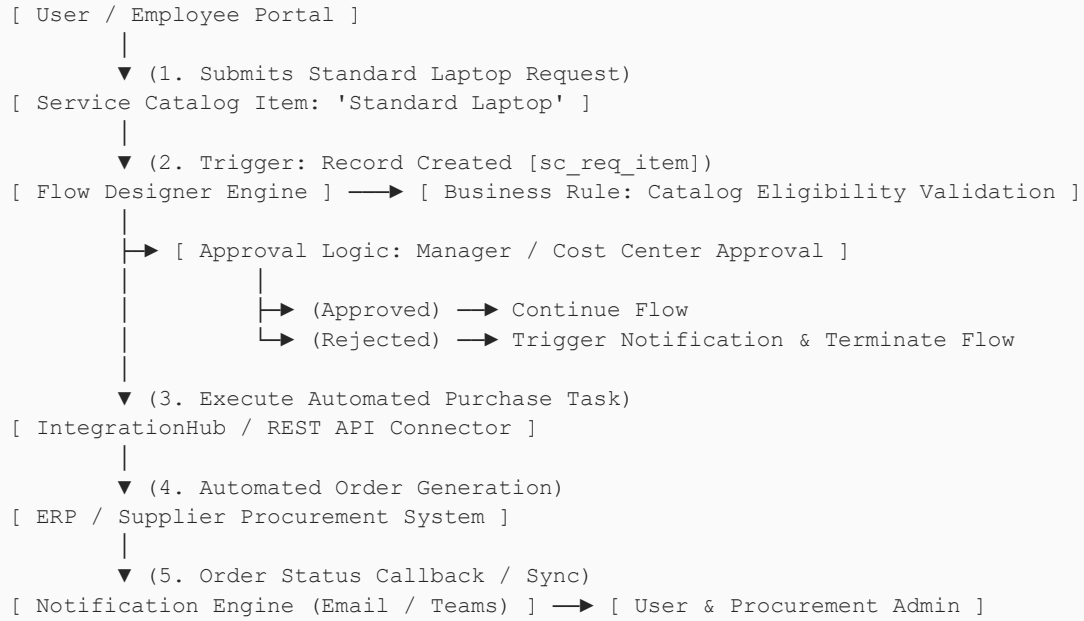
1. Executive Summary & Core Objectives

Solution architecture bridges the gap between business challenges and enterprise technology solutions. This document outlines the end-to-end architectural framework for transforming manually driven IT procurement into an automated, low-touch workflow powered by ServiceNow Flow Designer.

- **Business-Tech Alignment:** Delivering an enterprise-grade automated solution to resolve manual approval bottlenecks and order delays.
- **System Structure & Behavior:** Defining component relationships across Service Catalog, Flow Designer engine, Approval Engine, and ERP/Procurement connectors.
- **Lifecycle Requirements:** Detailing sprint execution phases, field mapping rules, integration triggers, and security controls.
- **Operational Specifications:** Providing a repeatable governance specification for solution implementation, exception handling, and post-deployment monitoring.

2. Solution Architecture & Data Flow Diagram

FIGURE 1: End-to-End Technical Architecture & Component Interaction Flow



3. Component Architectural Specifications

Architecture Layer	Technology Component	Key Functional Responsibility
Presentation / Portal Layer	Service Portal / Service Catalog	Captures user laptop request, user parameters, shipping details, and auto-populates hardware specifications.
Business Logic & Workflow Layer	ServiceNow Flow Designer Engine	Processes intake payload, executes business validation, manages approval Routing, and coordinates execution stages.
Security & Governance Layer	Approval Engine & ACLs	Enforces cost-center thresholds, handles manager routing, and restricts system field modifications to authorized roles.
Integration Layer	IntegrationHub / REST API Spoke	Transmits structured XML/JSON order payloads to third-party vendor systems or internal ERP platforms.
Data Storage Layer	CMDB & Procurement Tables	Stores catalog specifications, tracks order lifecycle statuses, updates asset inventory, and records audit logs.
Notification & Output Layer	Notification Engine	Sends automated email and messaging updates for approvals, order placements, and delivery tracking.

4. End-to-End Data Flow & Sequence

Step 1: Intake & Validation

User selects standard laptop from Service Catalog. System auto-populates specifications, checks user eligibility against active directory policy, and submits item.

Step 2: Flow Execution Trigger

A record creation on `sc_req_item` triggers the Flow Designer sequence 'Standard_Laptop_Procurement_Flow'.

Step 3: Approval Processing

Flow evaluates cost threshold. If > $0, routes approval task to line manager. Automated timeout/escalation triggers after 48 hours.

Step 4: Procurement Order Automation

Upon approval, Flow executes REST API callout to Procurement ERP to auto-generate Purchase Order (PO) and vendor fulfillment ticket.

Step 5: Status Tracking & Closure

Fulfillment webhook updates ServiceNow record status to 'In Fulfillment', triggers confirmation notification to employee, and updates asset inventory log.

5. Non-Functional Architecture Considerations

Architectural Dimension	Non-Functional Specification
Security & Access Control	Role-Based Access Control (RBAC) enforced via ServiceNow ACLs. Integration endpoints secured using OAuth 2.0.
Performance & SLA	Flow Designer execution complete within < 3 seconds per trigger. Approval notifications dispatched within 60 seconds.
Exception & Fallback	Failed API calls auto-retry 3 times before routing an exception task to IT Procurement Admins (USN12).
Scalability & Audit	Supports up to 10,000 requests/month without platform degradation. Full audit trail maintained for compliance.